Selecting the Right Flowmeter—Part 1

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("Choosing the Best Flowmeter")

With the many flowmeters available today, choosing the most appropriate one for a given application can be difficult. This article discusses six popular flowmeter technologies, in terms of the major advantages and disadvantages of each type, describes some unique designs, and gives several application examples.

Dozens of flowmeter technologies are available. This article covers six flowmeter designs—variable-area, mass, Coriolis, differential-pressure, turbine, and oval-gear. Table 1 compares the various technologies.

Table 1
A Comparison of Flowmeter Options

Attribute	Variable-area	Coriolis	Gas mass-flow	Differential-Pressure	Turbine	Oval Gear
Clean gases	yes	yes	yes	yes	yes	—
Clean Liquids	yes	yes	—	yes	yes	yes
Viscous Liquids	yes (special calibration)	yes	—	no	yes (special calibration)	yes, >10 centistokes (cst)
Corrosive Liquids	yes	yes	—	no	yes	yes
Accuracy, ±	2-4% full scale	0.05-0.15% of reading	1.5% full scale	2-3% full-scale	0.25-1% of reading	0.1-0.5% of reading
Repeatability, ±	0.25% full scale	0.05-0.10% of reading	0.5% full scale	1% full-scale	0.1% of reading	0.1% of reading
Max pressure, psi	200 and up	900 and up	500 and up	100	5,000 and up	4,000 and up
Max temp., °F	250 and up	250 and up	150 and up	122	300 and up	175 and up
Pressure drop	medium	low	low	medium	medium	medium
Turndown ratio	10:1	100:1	50:1	20:1	10:1	25:1
Average cost*	$200-600	$2,500-5,000	$600-1,000	$500-800	$600-1,000	$600-1,200

*Cost values can vary quite a bit depending on process temperature and pressures, accuracy required, and approvals needed.

Variable-Area Flowmeters

Design overview: The variable-area flowmeter (Figure 1) is one of the oldest technologies available and arguably the most well-known. It is constructed of a tapered tube (usually plastic or glass) and a metal or glass float. The volumetric flowrate through the tapered tube is proportional to the displacement of the float.



Figure 1
The plastic or glass tube of the variable-area flowmeter lets the user visually inspect the float, whose position in the tapered tub is proportional to the volumetric flowrate.

Fluid moving through the tube form bottom to top causes a pressure drop across the float, which produces an upward force that causes the float to move up the tube. As this happens, the cross-sectional area between the tube walls and the float (the annulus) increases (hence the term variable-area).

Because the variable-area flowmeter relies on gravity, it must be installed vertically (with the flowtube perpendicular to the floor). Some variable-area meters overcome this slight inconvenience by spring loading the float withing the tube (Figure 2). Such a design can simplify installation and add operator flexibility, especially when the meter must be installed in a tight physical space and a vertical installation is not possible.

Two types of variable-area flowmeters are generally available: direct-reading and correlated. The direct-reading meter allows the user to read the liquid or gas flowrate in engineering units (i.e., gal/min and L/min) printed directly on the tube, by aligning the top of the float with the tick mark on the flowtube.

The advantage of a direct-reading flowmeter is that the flowrate is literally read directly off the flowtube. Correlated meters, on the other hand, have a unitless scale (typically tick marks from 0 to 65, or 0 to 150), and come with a separate data sheet that correlates the scale reading on the flowtube to the flowrate in a particular engineering unit. The correlation sheets usually give 25 or so data points along the scale of the flowtube, allowing the user to determine the actual flowrate in gal/min, L/min, or whatever engineering unit is needed.

The advantage of the correlated meter is that the same flowmeter can be used for various gases and liquids (whose flow is represented by different units) by selecting the appropriate correlation sheets, where additional direct-reading meters would be required for different fluid applications. Similarly, if pressure or temperature parameters change for a given application, the user would simply use a different correlation sheet to reflect these new parameters. By comparison, for a direct-reading meter, a change in operating parameters will compromise the meter's accuracy, forcing it to be returned to the factory for recalibration. In general, the average accuracy of a variable-area flowmeter is ±2-4% of fullscale flow.

Advantages: The major advantage of the variable-area flowmeter is its relative low cost and ease of installation. Because of its simplicity of design, the variable-area meter is virtually maintenance-free and, hence, tends to have a long operating life.

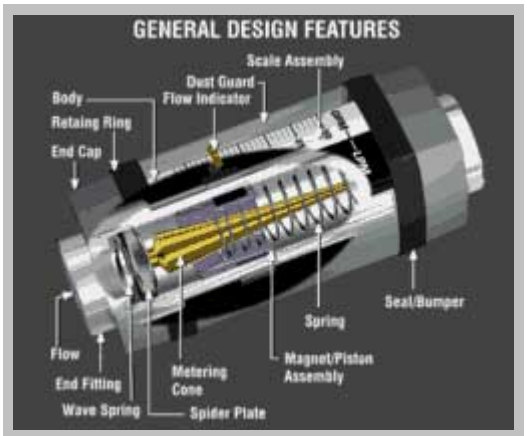


Figure 2
This variable-area meter with a spring-loaded float can be installed at any angle. This accommodation is not available for traditional variable-area flowmeters, whose operation relies on gravity.

Another advantage is its flexibility in handling a wide range of chemicals. Today, all-Teflon meters are available to resist corrosive damage by aggressive chemicals. The advantage of a Teflon flowmeter with a built-in valve is that you can not only monitor the fluid flowrate, but you can control it, as well, by opening and closing the valve. If the application requires an all-Teflon meter, chances are the fluid is pretty corrosive, and many users would like the option of controlling the flowrate by simply turning a valve that is built into the flowmeter itself.

Disadvantages: One potential disadvantage of a variable-area flowmeter occurs when the fluid temperature and pressure deviate from the

calibration temperature and pressure. Because temperature and pressure variations will cause a gas to expand and contract, thereby changing density and viscosity, the calibration of a particular variable-area flowmeter will no longer be valid as these conditions fluctuate. Manufacturers typically calibrate their gas flowmeters to a standard temperature and pressure (usually 70°F with the flowmeter outlet open to the atmosphere, i.e., with no backpressure).

During operation, the flowmeter accuracy can quickly degrade once the temperatures and pressures start fluctuating from the standard calibration temperature and pressure. Meters used for water tend to show less variability, since water viscosity and density changes very little with normal temperature and pressure fluctuations. While there is a way to correlate the flow from actual operating conditions back to the calibration conditions, the conventional formulas used are very simplified, and don't take into account the effect of viscosity, which can cause large errors.

Table 2
The Effect of Pressure Deviations on a Variable-Area Flowmeter

Maximum flowrate, L/min	Fluid temperature, °F	Outlet pressure, psi
Fluid type: Air		
2.23	70	0
1.65	70	15
1.30	70	35
2.26	90	0
2.28	110	0
2.32	150	0
Fluid type: water		
4.82	70	0
4.82	70	15
4.82	70	35
4.86	90	0
4.89	110	0
4.95	150	0

As Table 2 shows, the effect of pressure deviations can be quite significant. This table was created using data from a variable-area flowmeter that was calibrated for air at 70°F and with the outlet of the flowmeter vented to the open atmosphere (i.e. , 0 psi of outlet pressure).

The flowmeter was calibrated to read a maximum of 2.23 L/min at this temperature and pressure. When the outlet pressure increases as all other parameters remain constant, the flowrate drops off. This pressure change affects the viscosity and density of the gas and will cause the actual flowrate to deviate from the theoretical, calibrated flowrate. This relationship is extremely important to be aware of, and underscores the difficulty in measuring gas flow. Also note that even though gas flowrate changes with a change in gas temperature (with all other parameters remaining constant), this effect is much less significant with air than with other gases.

Table 2 shows this same variation with a meter calibrated for water at 9 psi venting pressure and a temperature of 70°F. Here, one can assume water to be incompressible. As shown, there is no direct effect on water flow with a change in back-pressure. The temp-erature change is not that significant either. But, for various fluids, a change in temperature could change the viscosity enough to degrade the accuracy below acceptable limits.

The bottom line is that the user must be aware of any variation between calibration conditions and operating conditions for gas flows, and must correct the reading according to the manufacturer's recommendations. Some users have the manufacturer calibrate the meter to existing conditions, but this presumes that operating conditions will remain the same—which they rarely do.

The effect of viscosity changes is another potential disadvantage of the variable-area meter when measuring liquids. When a viscous liquid makes its way through a variable-area flowmeter,

drag layers of fluid will build up on the float. this will cause a slower-moving viscous liquid to yield the same buoyant force as a faster-moving fluid of lower viscosity. The larger the viscosity, the higher the error. The general rule of thumb is as follows—unless the meter has been specifically calibrated for a higher-viscosity liquid, only water-like liquids should be run through a variable-area flowmeter.

Sometimes, for liquids that are slightly thicker than water, a manufacturer-supplied correction factor can be used without the need to recalibrate the whole meter. As always, check with the manufacturer if you plan on deviating from its calibration fluid and calibration conditions. For a more-detailed discussion of the proper correction equations to apply to variable-area flowmeters in both water and gas service when they deviate from standard conditions, consult Refs. 9 and 10.

Applications:
Variable-area flowmeters are well suited for a wide variety of liquid and gas applications, including the following:

- Measuring water and gas flow in plants or labs
- Monitoring chemical lines
- Purging instrument air lines (i.e., lines that use a valved meter)
- Monitoring filtration loading
- Monitoring flow in material-blending applications (i.e., lines that use a valved meter)
- Monitoring hydraulic oils (although this may require special calibration)
- Monitor makeup water for food & beverage plants

Mass Flowmeters

Design Overview: Mass flowmeters are one of the most popular gas-measurement technologies in use today (Figure 3). Most thermal mass flowmeters for gases are based on the following design principles, which are shown in Figure 4. a gas stream moves into the flowmeter chamber and is immediately split into two distinct flow paths. Most of the gas will go through a bypass tube, but a fraction of it goes through a special capillary sensor tube, which contains two temperature coils.



Figure 3
Because the mass flowmeter measures mass flow rather than volumetric flow, this popular device is relatively undaunted by fluctuations in line pressures and temperatures, especially compared with a variable-area flowmeter. The unit shown provides an integral digital display, as well as a built-in control valve.

Heat flux is introduced at two sections of the capillary tube by means of these two wound coils. When gas flows through the device, it carries heat from the coils upstream to the coils downstream. The resulting temperature differerential creates a proportional resistance change in the sensor windings.

Special circuits, known as Wheatstone bridges, are used to monitor the instantaneous resistance of each of the sensor windings. The resistance change, created by the temperature differential, is amplified and calibrated to give a digital readout of the flow.

As shown in Figure 3, the mass flowmeter is available with a built-in valve for flow-control applications. This allows for external control and the programming of a setpoint for a critical flowpoint. Most mass flowmeters also have an analog or digital output signal to record the flowrate. The average mass flowmeter has an accuracy of ±1.5-2% of fullscale flow.

Advantages: The main advantage of a mass flowmeter for gas streams is its ability (within limitations) to "ignore" fluctuating and changing line temperatures and pressures. As mentioned above for variable-area flowmeters, fluctuating temperatures and pressures will cause gas density to change, yielding significant flow errors. Because of the inherent design of the mass flowmeter, this problem is much less significant than that found in variable-area flowmeters. Mass flowmeters measure the mass or molecular flow, as opposed to the volumetric flow. One can think of the mass flowrate as the volumetric flowrate normalized to a specific temperature and pressure.

A more intuitive way to understand mass versus volumetric measurement is to imagine a gas-filled ballon. Although the volume of the balloon may be altered by squeezing it (changing the gas pressure), or by taking the balloon into a hot or cold environment (changing the gas temperature), the mass of the gas contained inside the balloon remains constant. So it is with mass flow as opposed to volumetric flow.

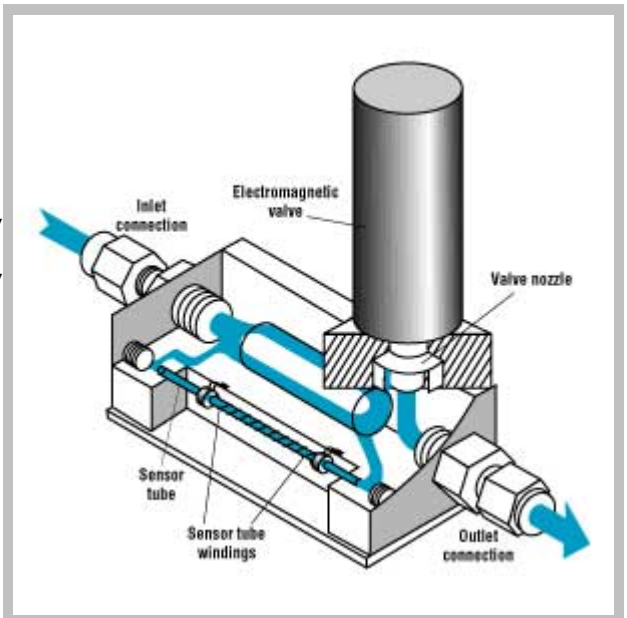


Figure 4
Inside a mass flowmeter, the gas is split. Most goes through a bypass tube, while a fration goes through a sensor tube containing two temperature coils. Heat flux is introduced at two sections of the sensor tube by means of two wound coils. As gas flows through the device, it carries heat from the upstream, to the downstream, coils. The temperature differential, generates a proportional change in the resistance of the sensor windings. Special circuits monitor the resistance change, which is proportional to mass flow, and calibrate it to give a digital readout of the flow.

A variable-area flowmeter measures volumetric flow. The flowrate on the flowtube reflects the volume of gas passing from the inlet to the outlet. This volume can change when gas temperatures and pressures change. Because a mass flowmeter is measuring the actual mass of gas passing form inlet to outlet, there is very little dependence on fluctuating temperatures and pressures. If you were piping an expensive gas, you would certainly want to keep track of the amount of gas used based on mass, not volumetric, flow.

Makers of mass flowmeters measure their products' ability to withstand changing pressures and temperatures by giving coefficients that state the deviation of accuracy per degree or psi change. For example, typical coefficient values are 0.10% error per degree C, and 0.02% error per psi. This means that each degree or psi change away from the meter's calibration conditions will degrade the accuracy by these coefficient amounts. So, although there is a dependence on pressure and temperature for a mass meter, its is very small, if not negligible. This is the biggest advantage of a mass flowmeter. Another is that there are no moving parts to wear out.

Disadvantages: Aside from the fact that the gas going through the mass flowmeter should be dry and free from particulate matter, there are no major disadvantage to the mass flow technology. Mass flowmeters must be calibrated for a given gas or gas blend.

Applications:
Applications for mass flowmeters are diverse, but here are some typical uses:

- Monitoring and controlling air flow during gas chromatography
- Monitoring CO_2 for food packaging
- Gas delivery and control for fermenters and bioreactors
- Leak testing
- Hydrogen flow monitoring (e.g., in the utility industry)
- Control of methane or argon to gas burners
- Blending of air into dairy products
- Regulating CO_2 injected into bottles during beverage production

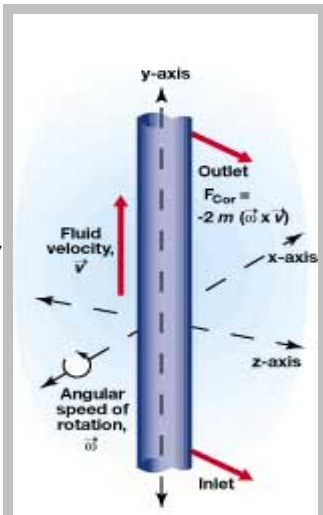
- Nitrogen delivery and control for tank blanketing

Coriolis Flowmeters

Design Overview: The Coriolis flowmeter is named for the Coriolis effect, an inertial force discovered by 19th-century mathematician Gustave-Gaspard Coriolis. as a result of the Coriolis force, the acceleration of any body moving at a constant speed with respect to the Earth's surface will be deflected to the right (clockwise) in the northern hemisphere, and to the left (counter-clockwise) in the southern hemisphere.

The basic design of the Coriolis meter makes use of this Coriolis force by subjecting a set of curved measuring tubes to rotary oscillations about an axis. This oscillation is normally driven by two electromagnetic coils, which also physically couple the two curved measuring tubes. As a particular fluid flows through the tubes, it will move through points of high rotational velocity, to points of lower rotational velocity.

Upon approaching the tube plane in which the rotational axis is located, the rotational motion of the fluid element is decelerated at a uniform rate, until it finally reaches zero in the plane of the rotational axis. As the fluid element flows away form the rotational axis plane, toward points with higher rotational velocity, it is uniformly accelerated to increasingly higher rotational velocities. This produces a force (the Coriolis force) that causes a twisting motion withing the sensor tubes (Figure 5a).



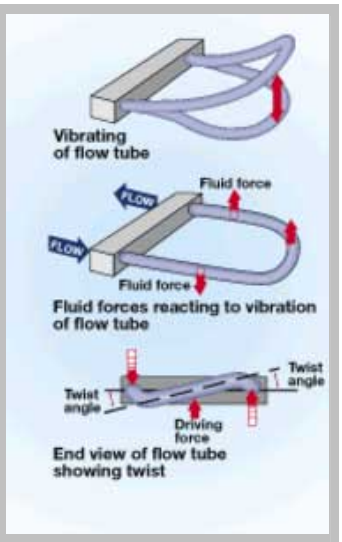


Figure 5a (left). In a coriolis flowmeter, the Coriolis force F_{Cor}, pushes out toward the z-axis as the fluid moves up through the tube. this force develops as the tube rotates at a rate of W around the x-axis, and causes the tube to distort out of the x-y plane

Figure 5b (right). As an example of a single-tube Coriolis flowmeter, this figure shows the fluid forces that generate the twisting motion of the flow tube

If v is the velocity of the fluid in the measuring tube, m/s, w the instantaneous angular speed of rotation, radians/s, and m the mass of liquid in the tube section, kg, then the following applies to the Coriolis force, kg(m/s) (Note that if the flow is low, you may be using different units to represent smaller forces):

$$F_{Cor} = -2m(w \times v)$$

The design of the Coriolis flowmeter takes advantage of this force in the following manner. First, the electromagnetic drivers initiate a vibration or oscillation in the sensor tube. This oscillation occurs even when there is no fluid moving in the meter.

The amplitude and frequency of this oscillation varies from manufacturer to manufacturer, but in general, the amplitude is about 3 millimeters, and the frequency is roughly 75-100 cycles/s. As the fluid element passes through the sensor tubes, the Coriolis forces come into play. The Coriolis forces cause a twisting, or distortion, in the measuring tube, which causes a vibrational phase difference between the two tubes.

Some designs use only one sensor tube (figure 5b). In this case, the distortion caused by the Coriolis force in the tube is compared to the tube at "no flow" conditions. In both cases, however, a correlation to the mass flowrate is achieved, because the measured phase difference or distortion is directly proportional to the mass flowrate of the fluid. Meanwhile, temperature-compensation techniques nullify the temperature dependence of the tube oscillations, creating a high-accuracy correlation to mass flow.

Advantages: The biggest advantage of the Coriolis design is that it measures mass flow instead of volumetric flow. Because mass is unaffected by changes in pressure, temperature, viscosity and density, reasonable fluctuations of these parameters in the fluid line have no affect on the accuracy of the meter, which can approach 0.05% of mass flow.

Coriolis meters can also determine fluid density by comparing the resonant frequency of the fluid being measured with that of water. Knowing density, the software can then convert mass to volume or percent solids.

Since there are no obstructions in the fluid path, Coriolis meters have inherently low pressure drop for low-viscosity liquids. Turndown ratios (the ratio of maximum to minimum flow) of 100:1 are not uncommon. In addition, the lifetime and reliability of the Coriolis meter are high as the flow path is free of moving parts and seals. And, if installed properly, vertically installed Coriolis meters are self draining, so they will not hold fluid when the line is down. A variety of wetted parts, communications outputs and connections are available.

Disadvantages: Because of their high accuracy and reliability, Corilois meters tend to be relatively expensive. This is not necessarily a disadvantage, however, if one looks at the relatively low cost of installation and ownership over time (Table 1). Because of their accuracy, Coriolis meters can help increase operating efficiency and save on production costs.

The main limitation of the Coriolis meter is that pressure drop can become large as fluid viscosity increases. For viscous products, check with the manufacturer to make sure the pressure drop at you max flowrate is acceptable and within your design parameters.

Applications:
Coriolis flowmeters are suitable for:

- General-purpose gas or liquid flow
- Custody transfer
- Monitoring concentration and solids content
- Blending ingredients and additives
- Conducting a primary check on secondary flowmeters
- Metering natural-gas consumption
- Monitoring such fluids as syrups, oils, suspensions and pharmaceuticals

Differential-Pressure Meters

Design overview: While many different types of differential-pressure flowmeters are available, this discussion will focus on one type. The technology discussed here involves the measurement of a pressure differential across a stack of laminar flow plates (Figure 6). During operation, a pressuredrop is created as fluid enters through the meter's inlet. The fluid is forced to form thin laminar streams, which flow in parallel paths between the internal plates separated by spacers.

The pressure differential created by the fluid drag is measured by a differential-pressure sensor connected to the top of the cavity plate. The differential pressure from one end of the laminar flow plates to the other end is linear and proportional to the flowrate of the liquid or gas.

What makes this technology unique is the linear relationship between differential pressure, viscosity and flow, which is given by the following equation

$$\mathbf{Q = K[P_1\text{-}P_2)/n_2]}$$

where (units vary per approach):
Q = Volumetric flowrate
$\mathbf{P_1}$ = Static pressure at the inlet
$\mathbf{P_2}$ = Static pressure at the outlet
n = Viscosity of the fluid
K = Constant factor determined by the geometry of the restriction

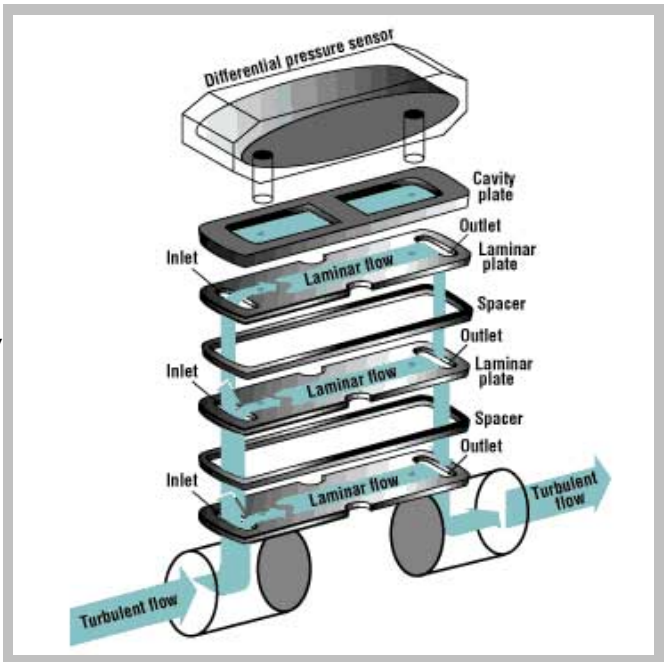


Figure 6
Using a differential-pressure flowmeter, a pressure drop is created as fluid enters the inlet. The fluid is forced to form thin laminar streams, which flow in along parallel plates. The pressure differential created by fluid drag from one end of the laminar flow plates to the other is linear and proportional to the flowrate of the liquid or the gas.

This direct relationship between pressure, viscosity and flow allows the meter to switch easily among different gases without recalibration. This is normally accomplished by programming in the various gas viscosities and allowing the user to dial in the appropriate gas, via a set of switches.

Variances in temperature and pressure, which often cause errors in variable-area flowmeters, can be easily handled by adding a pressure sensor (separate form the differential-pressure sensor in the basic design) and a temperature sensor to the design, to constantly monitor fluctuations in stream pressure and temperature, and correct the flow readings to standard pressure and temperature (77°F and 1 atm). This is critical for gas flowmeters, which are very sensitive to these parameters. Typical accuracy for the design is ±2-3% fullscale.

Advantages: As with mass flowmeters, the differential-pressure meter has no moving parts to wear out. And, unlike with mass flowmeters, users of differential-pressure meters can measure different gases, such as air, hydrogen, ethane, methane, nitrous oxide, carbon dioxide, carbon monoxide, helium, oxygen, argon, propane and neon, by setting a switch on the unit, without the need for recalibration.

For control applications, these meters are available with a built-in proportioning valve for onboard or remote control of the flowrate. With a wide variety of flow ranges and models for both gases and liquids, the differential-pressure meter is one of the most versatile designs currently on the market.

Disadvantages: These meters are generally reserved for use with clean gases and liquids. particulates with diameters >20 to 30 micrometers could get caught between the plates.

Applications:
Viable applications include the following:

- Chemical applications (ratio, metering, and additive control)
- Pharmaceutical applications (liquid injection and batching)
- Research and development, and laboratory applications (gas blending, injection and aeration)
- Food and beverage applications (CO_2 measurements, air drying, and process control)

Turbine Meters

Design Overview: Many designs exist for turbine flowmeters, but most are a variation on the same theme. As fluid flows through the meter, a turbine rotates at a speed that is proportional to the flowrate (Figure 7). Signal generators, usually located within the rotor itself, provide magnetic pulses that are electronically sensed through a pickup coil (the yellow pickup coil shown in Figure 7) and calibrated to read flow units. In some designs, an integral display may show both the flowrate and the total flow since power-up. Turbine meters are available for both gas and liquid flow.

Because of the rotating blades in a turbine meter, the output signal will be a sine wave voltage (V) of the form:

V=KwsinNwt

where:
K = The amplitude of one sine wave
w = The rotational velocity of the blades
N = The number of blades that pass the pickup in one full rotation
t = Time

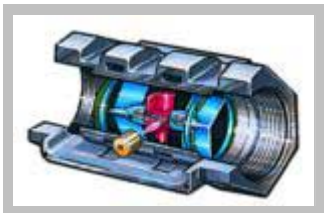


Figure 7
This cutaway view of a turbine flowmeter shows the turbines and signal generators used to produce voltage pulses that are proportional to the flowrate.

Because the output signal is proportional to the rotational velocity of the turbines—which, in turn, is proportional to the liquid flow—the signal is easily scaled and calibrated to read flowrate and flow totalization. Turbine flow sensors generally have accuracies in the range of ±0.25-1% fullscale.

Advantages: The main advantages of the turbine meter are its high accuracy (±0.25% accuracy or better is not unusual) and repeatability, fast response rate (down to a few milliseconds), high pressure and temperature capabilities (i.e., up to 5,000 psi and 800°F with high-temperature pick coils), and compact rugged construction. Some manufacturer's have taken turbine meter design to the next level by incorporating advanced electronics that perform temperature compensation, signal conditioning and linearization, all within a few milliseconds. This advanced technology will allow the meter to automatically compensate for viscosity and density effects.

Disadvantage: The disadvantage of the turbine meter is that is relatively expensive and has rotating parts that could clog from larger suspended solids in the liquid stream. And, most turbine meters need a straight section of pipe upstream from the flowmeter in order to reduce turbulent flow. This may make installation a challenge in small areas. However, some newer turbine meters reduce or eliminate the amount of straight pipe required upstream, by incorporating flow straighteners into the body of the unit.

Another disadvantage in some designs is a loss of linearity at the low-flow end. Low-velocity performance and calibration can be affected by the natural change in bearing friction over time. However, today's self-lubricated retainers, low-drag fluid bearings, and jeweled-pivot bearings all help to reduce the friction points, thereby allowing for greater accuracy and repeatability in lower-flow applications.

Applications:
Turbine flowmeters can be found in a wide variety of industries and applications:

- Rotometer replacement
- Pilot plants
- Research and development facilities
- Cooling water monitoring
- Inventory control
- Test stands
- Water consumption
- Makeup water

Oval-Gear Flowmeters

Design Overview: The design of the oval-gear flowmeter is relatively simple: oval-shaped, gear-toothed rotors rotate within a chamber of specified geometry (Figure 8). As these rotors turn, they sweep out and trap a very precise volume of fluid between the outer oval shape of the gears and the inner chamber walls, with none of the fluid actually passing trough the gear teeth. Normally, magnets are embedded in the rotors, which then can actuate a reed switch or provide a pulse output via a specialized, designated sensor (such as a Hall Effect sensor). Each pulse or switch closure then represents a precise increment of liquid volume that passes through the meter. The result is a high accuracy (usually ±0.5 percent of reading) and resolution, and almost negligible effects for varying fluid viscosity, density and temperature.



Figure 8
During operarion, each gear rotation in the oval-gear meter traps a pocket of fluid between the gear and the outer chamber walls. A designated sensor counts the pockets of fluids passing from inlet to outlet, and correlates this value to a flowrate.

When sizing an oval-gear flowmeter, keep in mind that the higher the fluid viscosity, the more pressure will be required to "push" the fluid into the flowmeter and around the gears. Essentially, the pressure drop is the only limiting factor when the application requires the metering of highly viscous liquids.

The general rule is that as long as the fluid will flow, and as long as there is enough system pressure, the oval-gear meter will be able to measure the flow. In applications where the lowest possible pressure drop is required, some manufacturers can replace the standard rotors with specially cut, high-viscosity rotors. The manufacturer will be able to provide a graph of flowrate versus pressure drop for various viscosities.

The oval-gear flowmeter works best when there is a little backpressure in the line; a throttling valve on the meter outlet usually works just fine. The oval-gear meter is not suitable for gases, including steam and multi-phase fluids.

Advantages: The advantage of the oval-gear flowmeter is the it is, withing certain limits, largely independent of the fluid viscosity (users should just remain aware that higher pressures will be required to push higher-viscosity fluids through the meter). This opens up a whole range of applications, including the metering of oils, syrups and fuels.

Ease of installation is another advantage of th oval design. Because no straight pipe runs or flow conditioning is required, these meters can be installed in tight areas, allowing for more flexibility in application design.

Disadvantage: Oval-gear meters are generally not recommended for water or water-like fluids, because the increased risk of fluid slippage between the gears and chamber walls. Fluid slippage will cause a slight degradation in accuracy, with low-viscosity fluids being more prone to degradation. As viscosity increases, the wall slippage quickly becomes minimal, and the best accuracy is realized. Since the oval-gear meter is really designed for higher-viscosity fluids, it can be argued that running water through them is not a viable application anyway.

Applications:
Oval meters are best suited for the following applications:

- Measurement of net fuel use in boilers and engines
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- Verification of proper bearing-lubricant delivery in hydraulic applications
- Monitoring of paper-finishing chemicals
- Monitoring the flow of wax finishes
- Monitoring syrup injection in main beverage lines
- Monitoring and batching volumes of thick candy coating
- Monitoring and automating the dispensing of cooking oils

The specifications for the six flowmeter designs discussed above will vary widely from manufacturer to manufacturer, and the performance values provided represent an average. When selecting a flowmeter for a given attribute, the engineer should consider additional attributes—including velocity-profile deviations, the effect of non-homogeneous or pulsating flow, and cavitation, all of which will affect flowmeter choice, installation and operation. While beyond the scope of this article, a thorough discussion of these parameters can be found in Ref. 5.

Selecting the Right Flowmeter—Part 2

Use pros and cons to select from these unique flowmeter technologies!

In this article, five flow-measurement technologies are summarized: bubble, Doppler, transit-time, vortex, and magnetic.

After reviewing the basic design parameters and highlighting the pros and cons associated with each flowmeter type, process applications for each technology will be discussed. The information is then summarized at the end of this article in a table (Table 1: A Comparison of Flowmeter Parameters), which compares the various attributes of these five technologies, such as accuracy, maximum pressures and temperatures, and average costs. The intention of this article is not to recommend a flowmeter for every possible application, but rather to provide the basic knowledge needed to make an informed flowmeter selection among these types for a given application.

The Bubble Flowmeter

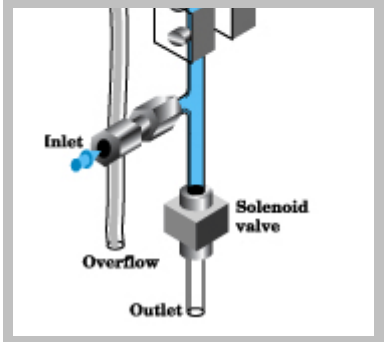
The bubble flowmeter is not as well known as other types. This is unfortunate, since the bubble meter offers some features not found in more-expensive and more-intricate designs.

Design Overview: Historically, the bubble meter has found its niche in the field of gas-chromatography analysis where it is used to measure column, detector, and carrier-gas flowrates. Today, however, the bubble meter is available in a larger variety of flow ranges for both liquids and gases, which greatly increases the number of potential applications.

Although there are manual bubble meters that require timing of the bubble movement with a stopwatch and referencing from a printed flowrate chart, this discussion focuses on the more-sophisticated electronic flowmeters that give a digital readout without operator involvement. There are two general designs to a bubble meter; the designs are distinctly different for gases and for liquids.

The bubble meter design for liquids makes use of a timed measurement of a meniscus rising between two optical sensors (Figure 1). In order to understand how this technology is able to measure the volumetric flowrate, one may follow the fluid path inside the flowmeter from the beginning to the end. First, fluid enters the inlet and moves up inside the glass tube, past the sensor block and around the tube toward the outlet. As this happens, the solenoid valve is timed to periodically open and close, thereby sucking a small amount of air into the tube. This creates separate columns of liquid that move upward inside the tube, and toward the optical-sensor block. The meniscus that is formed by these columns of fluid against the glass capillary-tube walls is measured by the optical sensors. Since the meniscus travels at the same rate as the column of fluid, measuring the rate of meniscus-travel gives a direct correlation to the liquid flow.

Figure 1
In a liquid-bubble meter, the speed of the meniscus created by the air gap is measured within the optical sensor block. The elapsed time for the meniscus to pass between the lower and upper sensor block is proportional to the volumetric flowrate.



Two infrared sensors located within the sensor block time the rise rate of the meniscus, and this volume-over-time measurement is then converted to a flowrate and displayed on

a digital readout. As the fluid moves around the top of the tube, air is vented at the top while the liquid continues around and exits at the overflow tube. The process then repeats itself as the solenoid valve opens to create another air gap.

By comparison, the bubble design for gas flow works a little differently although the same basic concept remains (Figure 2). For the gas bubble flowmeters, a soapy solution is used to fill the lower reservoir of the glass flow tube. The gas flow source is then connected to a point above the bubble-solution reservoir and gas travels around to the glass flow tube. At this point, the rubber bulb is either manually squeezed or a clamp is used to continuously generate bubbles that travel at the same speed as the gas.



Figure 2
The gas-bubble meter works very similarly to the liquid-bubble meter, but instead of a liquid meniscus, a bubble is created in the flow stream, and it is the speed of the bubble that is timed between the sensor blocks.

When the bubble passes the lower optical sensor within the sensor block, an internal timer is automatically started, and when the bubble passes the upper optical sensor, the timer is stopped. The total elapsed time is correlated to a gas flowrate and displayed on a digital readout. The small amount of liquid soap left over from the process collects in the flow trap (partially shown in the back of the unit) for disposal.

Advantages: The major advantage of the bubble meter for gases is that it is not affected by the gas composition. By contrast, most electronic meters must be calibrated for a specific gas or gas mixture. The traditional gas mass flowmeter is a good example of this. A mass flowmeter calibrated for air will not work on other gases or gas mixtures without factory recalibration. When the gas is changed, the calibration must be updated.

This is not the case with a bubble flowmeter. Whether one is measuring ordinary gases such as N_2, O_2, H_2, CO_2, and Ar, or measuring a unique gas mixture, one bubble meter can do it all. This versatility helps to lower equipment costs and can save recalibration time. Admittedly, it should be kept in mind that some gases may have a chemical reaction with he water used to make the bubble solution; the user should be careful when specifying bubble flowmeters for such compounds.

Another useful advantage of the bubble design is that the calibration does not drift over time. The main electrical parts of the system are the optical sensors for detecting the presence or absence of a bubble or meniscus layer. These noncontact sensors do not wear out or experience a drift in accuracy. The glass tube is fixed in diameter and will not change with time. Although we recommend returning the unit periodically for calibration service, don't be surprised if it is still well within the specified accuracy range.

In the gas-chromatography market, bubble meters can be qualified as a primary flow standard. Each unit can be individually calibrated to a U.S. National Institute of Standards and Technology (Gaithersburg, MD.; nist.gov) registered burette.

Traditionally only available for very low flowrates, bubble flowmeters are now available for expanded flowrate-ranges. While gas flows ranging from 0.1 to 25 L/min can be accurately measured, liquid bubble meters don't have quite the range as the gas versions and are available in sizes ranging from roughly 1 ml/min to 30 ml/min.

Disadvantages: In order to make an inline measurement with a bubble flowmeter, one needs to make a break in the line where the flow reading is desired, then make measurement and finally restore the line to its original condition. Bubble meters are therefore adequate for "end-of-line" readings, but are not well suited for continuous, in-line monitoring. In some applications, the use of a bubble solution could be a minor inconvenience, since it needs to be cleaned up after the measurement.

Applications: Bubble meters are most appropriately applied in laboratory and low flow research applications. Their use in more industrial applications is extremely limited. Some of the popular applications for a bubble flowmeter include:

- Supercritical fluid extraction
- Chromatography column, detector, and carrier-gas measurement
- Monitoring post detector flow volumes in HPLC systems
- Calibration and flow verification for variable area and electronic flowmeters
- Accurate flow measurement of gas mixtures without recalibration
- Accurate flow measurement of changing gas concentrations
- Calibration of air sampling pumps
- General purpose gas flow verification

The Doppler Flowmeter

Anyone that has heard the pitch of a train whistle change as the train passes has experienced the Doppler effect, named after the 19th century Austrian scientist Christian Doppler. This effect can be used to measure the flow in a pipe.

Design Overview: The Doppler effect is the frequency shift that occurs when a sound source (transmitter) is in relative motion with a receiver of that sound source. In the case of a Doppler flowmeter, we have two sensors mounted or strapped on the outside of a pipe. One of the sensors is the transmitter, and transmits a high frequency (ultrasonic) signal into the pipe. This signal is reflected off particulate matter or entrained gas bubbles in the fluid. The reflected signal is then picked up by the receiving signal and the frequency difference between the transmitted and reflected signals is measured and correlated into an instantaneous flowrate or flow total (Figure 3).

The frequency is subject to two velocity changes; one upstream and the other downstream. Traveling upstream, the velocity of the wave is given as (V_s - V cosθ) where V_s equals the velocity of sound in the fluid, V equals the average fluid velocity and θ equals the angle of the ultrasonic beam to the fluid flow. Similarly, the downstream velocity is given as (V_s + V cosθ). The Doppler relationship between the reflected and transmitted frequencies can now be expressed as:

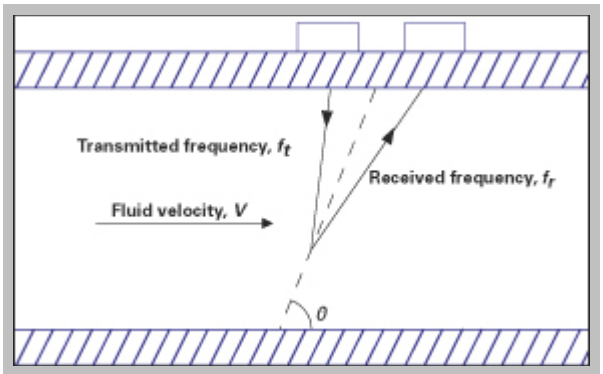


Figure 3
This illustration shows the Doppler signal penetrating the pipe and then reflecting off the particulates in the stream. The signal phase shift is measured and correlated to a flow velocity.

$$\mathbf{f_r = f_t[(V_s+V\ cos\theta)/(V_s - V\ cos\theta)]}$$

Here, f_r is the received frequency and f_t is the transmitted frequency. To further simplify this equation, one can assume that the velocity of the fluid in the pipe is much lower than the velocity of sound in the pipe; that is, $V << V_s$. With this assumption, one can write:

$$\mathbf{f_r = f_t[(V_s + V\ cos\theta)/V_s + (V\ cos\theta)]/V_s}$$

Which reduces further to

$$\mathbf{f_r = f_t[1 + (2V\ cos\theta)/V_s]}$$

The frequency shift is given by $\Delta f = f_r - f_t$ so that

$$\mathbf{\Delta f = [2(f_t)\ cos\theta/Vs]V}$$

Because ($2f_t\ cos\theta/V_s$) is a constant, one can write the final equation as

$$\mathbf{\Delta f = kV}$$

where

$$\mathbf{k = 2(f_t)\ \cos\theta / V_s}$$

This indicates that the fluid velocity in the pipe is directly proportional to the change in frequency between the transmitted and reflected ultrasonic signals. With knowledge of the pipe size, the electronics of the flowmeter will correlate the fluid velocity into a flowrate in the engineering unit of choice. Software corrections may have to be made for V_s, since the sound velocity through the medium will change with pressure and temperature fluctuations.

There are ultrasonic designs on the market that use a series of pulsed signals, as opposed to a continuous ultrasonic beam. The main advantage of the pulsed technology is that it can measure the vertical velocity profile within the pipe. Fluid flow will be faster along the middle of the pipe than along the pipe walls and the pulse-design allows one to obtain a better image the flow profile within the pipe.

Another sensor design that minimizes external noise uses dual-frequency Doppler technology to send two independent signals into the pipe at different frequencies. Since both signals are subject to the same Doppler shift, but the noise signals are random, the signals can be combined to calculate a flow velocity while subtracting out the noise.

Ultrasonic sensors can be used with a wide variety of pipe materials, but some will not allow the signal to pass through. Although pipe material recommendations will vary depending on the sensor design, you should not expect to have any problems with carbon steel, stainless steel, PVC, and copper. However, pipes made of concrete, fiberglass, iron, and plastic pipes with liners, could pose transmission problems. One should check with the particular manufacturer to ensure that the pipe material is suitable. Some Doppler designs utilize a section of pipe with built-in transducers that make direct contact with the fluid. This design, although no longer non-invasive, eliminates the problem of incompatible pipe materials.

The accuracy of the ultrasonic Doppler meter is typically around ±2% of full scale. Minimum concentration and particulate size required is roughly 25 PPM at 30 microns. Since some meters may require slightly larger concentrations, it is a good idea to check with the manufacturer. The vast majority of Doppler meters are used for liquids (roughly 88%) while the rest are used for gas (11%) and steam (1%) applications.

Advantages: The main advantage of the Doppler ultrasonic meter is its non-intrusive design. An acoustic-coupling compound is used on the surface of the pipe and the sensors are simply held in place to take a measurement or, for a more permanent installation, they are strapped around the pipe. Some manufacturers offer a special clamp-on probe which allows connection to smaller pipe sizes (down to 1/4-in. diameter). Other advantages include:

- Easy installation and removal—no process downtime during installation
- No moving parts to wear out
- Zero pressure drop
- No process contamination
- Works well with dirty or corrosive fluids
- Works with pipe sizes ranging from 1/2" to 200"
- No leakage potential
- Meters are available that work with laminar, turbulent, or transitional flow characteristics
- Battery powered units are available for remote or field applications
- Sensors are available for pulsating flows
- Advanced software and datalogging features available
- Insensitive to liquid temperature, viscosity, density or pressure variations

Disadvantages: Every flowmeter has its disadvantages and the Doppler design is no exception. The main disadvantage to the technology is the fact that the liquid stream must have particulates, bubbles, or other types of solids in order to reflect the ultrasonic signal. This means

that the Doppler meter is not a good choice for DI water or very clean fluids. Although strides have been made with the Doppler technology so that it can work with smaller particulate sizes and smaller concentrations, one still needs to have some particulates present (one design avoids this problem by placing a 90-deg. elbow a few pipe diameters upstream of the flow sensor, and sensing the turbulent swirls created by the elbow). A good rule of thumb is to have a bare minimum of 25 PPM at roughly 30 microns in order for the ultrasonic signal to be reflected efficiently. Some flowmeter designs may require a little more than this, so it is advisable to check the specifications of the meter one is considering.

Note that if the solids content is too high (around 50% and higher by weight), the ultrasonic signal may attenuate beyond the limits of measurability. This possibility should also be checked with the manufacturer, referring to one's specific application. Another disadvantage is that the accuracy can depend on particle-size distribution and concentration and also on any relative velocity that may exist between the particulates and the fluid. If there are not enough particulates available, the repeatability will also degrade.

Finally, the only other potential problem of this technology is that it can have trouble operating at very low flow velocities. If you suspect this may be a problem for an application, the low-end velocity that may be obtained with a particular sensor design should be checked with the manufacturer.

Applications: Doppler meters, being non-instrusive, have a wide variety of applications in the water, waste water, heating, ventilation and air conditioning (HVAC),HVAC, petroleum and general process markets. Below is a list of viable applications:

- Influent and effluent water flow
- Clarifier monitoring
- Digester feed control
- Waste water
- Potable water
- Cooling water
- Makeup water
- Hot and chilled water
- Custody transfer
- Water injection
- Crude-oil flow
- Mining slurries
- Acids
- Caustics
- Liquefied gases

The Transit-Time Flowmeter

Design Overview: Like its Doppler cousin, transit-time meters utilize an ultrasonic pulse that is projected into and across the pipe. The design works on a slightly different principle, however. The basic premise of the transit-time meter is to measure the time difference (or frequency shift) between the time of flight down-stream and the time of flight up-stream. This frequency shift can then be correlated into a fluid flowrate through the pipe. To help explain one type of transit-time design, Figure 4a shows two transducers attached to a pipe.

In this figure, V is the average fluid velocity, Z is the distance from the upstream transducer to the downstream transducer, and q is the angle between the ultrasonic-beam line and the horizontal fluid flow. The time it takes for the ultrasonic signal to go from the upstream transducer to the downstream transducer can be written as

$$t_{down} = Z/(V_s + V \cos\theta)$$

where V_s is the velocity of sound through the liquid. The upstream time can be written as (Figure 4b):

$$t_{up} = Z/(V_s - V \cos\theta)$$

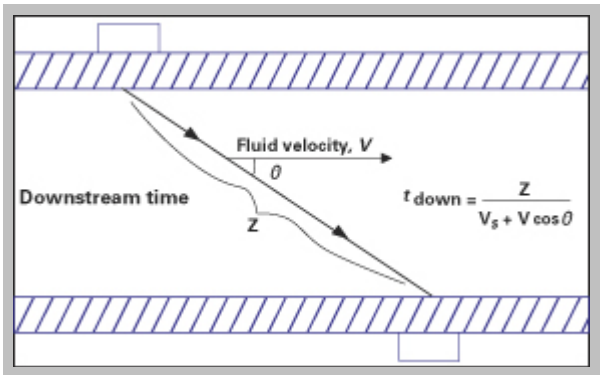


Figure 4a
This diagram of a transit-time flowmeter shows the downstream signal being projected between the two transit-time sensors.

Because the upstream and downstream frequencies can be generated in proportion to their respective transit-times, we can say the following:

$$f_{down} = 1/t_{down}$$

and

$$f_{up} = 1/t_{up}$$

where f_{down} and f_{up} represent the downstream and upstream frequencies respectively. The change in frequency can then be given as

$$\Delta f = f_{down} - f_{up} = 1/t_{down} - 1/t_{up}$$

By substitution, one obtains

$$\Delta f = (V_s + V \cos\theta)/Z - (V_s - V \cos\theta)/Z = (2 \cos\theta/Z)V$$

Since (2 cosθ/Z) is just a constant, one can write the final equation as

$$\Delta f = kV$$

with

$$k = 2 \cos\theta/Z$$

This, then, is the basic relationship used to determine flow velocity from the measured frequency shift. The flow rate can then be calculated using a Reynolds-number correction for velocity profile and by programming in the internal pipe diameter. The Reynolds-number correction takes into account the behavior of the fluid as being laminar, transitional or turbulent. These calculations are made electronically and the flowrate or flow total can then be displayed in the engineering units of choice. Interestingly enough in this instrument, the frequency shift is measured independently of V_s. This is an advantage, since corrections will not have to be made for the variance of V_s because of line-pressure and temperature fluctuations. Most transit-time applications

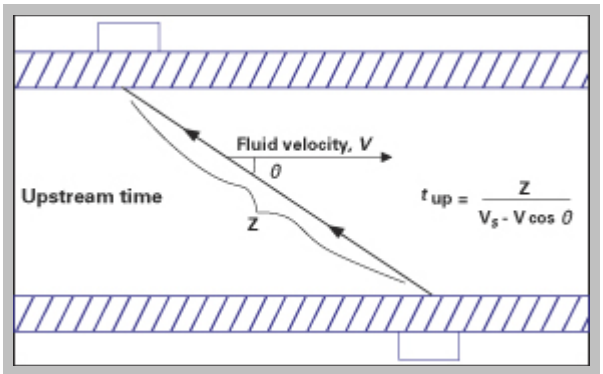


Figure 4b
This diagram shows the upstream signal projection. The frequency difference between the upstream and downstream times is proportional to the flow velocity.

involve liquids, but designs are available to handle gases, as well.

In light of the single path design discussed above, note that a single ultrasonic pulse will average the velocity profile across the transit path, and not across the pipe cross-section, where better accuracy would be obtained. Some flowmeters on the market send several ultrasonic pulses on separate paths in order to average this velocity profile; these meters tend to have better accuracy than their single-pulse counterparts. Transit-time flowmeters generally exhibit accuracies of around ±1% of the measured velocity. Pipe-material recommendations are the same as those given for Doppler flowmeters.

Advantages: As pointed out, the main advantage of the transit-time meter is that it works non-invasively with ultrapure fluids. This allows the user to maintain the integrity of the fluid while still measuring the flow. Some of the other advantages are listed below.

- Easy installation—transducer set clamps onto pipe
- No moving parts to wear out
- Zero pressure drop
- Can detect zero flow
- No process contamination
- Works well with clean and ultrapure fluids
- Works with pipe sizes ranging from 1" to 200"
- No leakage potential
- Meters available that work with laminar, turbulent, or transitional flow characteristics
- Battery powered units available for remote or field applications
- Sensors available for pulsating flows
- Advanced software and datalogging features available
- Insensitive to liquid temperature, viscosity, density or pressure variations

Disadvantages: Transit-time flowmeter performance can suffer from pipe-wall interference, and accuracy and repeatability problems can result if there are any air spaces between the fluid and the pipe wall. Concrete, fiberglass and pipes lined with plastic can attenuate the signal enough to make the flowmeter unusable. Because these factors can vary from one design to the next, it is advisable to check with the manufacturer to ensure that the pipe material is appropriate.

As mentioned before, the transit-time meters will not operate on dirty, bubbly, or particulate-laden fluids. Sometimes, the purity of a fluid may fluctuate so as to affect the accuracy of the flow measurement. For such cases, there are hybrid meters on the market that will access the fluid conditions within the pipe and automatically chose Doppler or transit-time operations where appropriate. These units are especially useful if the unit is to be used in a wide variety of different applications which may range from dirty to clean fluids.

Applications: Transit-time meters have wide applicability for flow measurement of clean or ultrapure streams. Some of these applications are listed below.

- Clean water flowrate in water treatment plants
- Hot or cold water in power plants, airports, universities, shopping malls, hospitals and other commercial buildings
- Pure and ultra-pure fluids in semiconductor, pharmaceutical, and the food & beverage industries
- Acids and liquefied gases in the chemical industry
- Light to medium crude oils in the petroleum refining industry
- Water distribution systems used in agriculture and irrigation
- Cryogenic liquids
- Gas-stack flow measurement in power plant scrubbers

The Vortex Flowmeter

Design Overview: At 11 a.m. on November 7th, 1940 the Tacoma Narrows suspension bridge in the state of Washington collapsed from wind-induced vibrations. The torsional motion of the bridge shortly before its collapse is an indication of the power of vortex shedding. The prevailing theory on the collapse of the bridge is that the oscillations were caused by the shedding of turbulent vortices in a periodic manner. Experimental observations have in fact shown that broad flat obstacles (also referred to as bluff bodies) produce periodic swirling vortices which generate high and low pressure regions directly behind the bluff body. The rate at which these vortices shed is given by the following equation:

f = SV/L

where,
f = the frequency of the vortices
L = the characteristic length of the bluff body
V = the velocity of the flow over the bluff body
S = Strouhal Number and is a constant for a given body shape

In the case of the Tacoma bridge, a wind speed of approximately 40 mph caused the formation of vortices around the 8-ft.-deep, steel plate girders of the bridge. This established vortices which were shed, according to the above equation, at approximately 1 Hz. As the structural oscillations constructively reinforced, the bridge began oscillating, building up amplitude, until it could no longer hold itself together.

Another less tragic example of the vortex principle can be seen in the waving motion of a flag. The flag pole, acting as a bluff body, creates swirling vortices behind it that give the flag its "flapping" quality in strong winds.

A practical application of vortex production can be found in the design of the vortex flowmeter. In this design, a bluff body or bodies is placed within the fluid stream. Just behind the bluff body, a pressure transducer, thermistor, or ultrasonic sensor picks up the high and low pressure and velocity fluctuations as the vortices move past the sensor (Figure 5). These fluctuations are linear, directly proportional to the flowrate and independent of fluid density, pressure, temperature and viscosity (within certain limits). As given explicitly in the above equation, the frequency of the vortices is directly proportional to the velocity of the fluid. Vortex meters are very flexible and the technology can be used for liquid, gas and steam measurements. This, along with the fact that they have no moving parts, makes them a very popular choice. Accuracies are typically in the ±1% range.

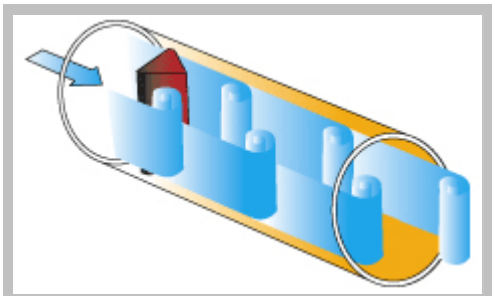


Figure 5
As fluid moves around the baffles, vortices form and move downstream. The frequency of the vortices is directly proportional to the flowrate.

Generally speaking, in-line vortex meters are available in line sizes ranging from 1/2 to 16". Insertion vortex meters that are installed in the top or sides of a pipe can be used for even larger pipe sizes. This makes them versatile in a wide variety of applications (Figure 6).

One final remark concerns the Reynolds number limitations for these flowmeters. For vortex meters, vortices will not be shed under a Reynolds number of approximately 2000. From roughly 2000 to 10,000, vortices will be shed but the resulting fluctuations are non-linear in this range. Typically, a minimum Reynolds number of 10,000 is required in order get optimum performance from the vortex flowmeter. This number can vary from one design to another, so it is advisable to check with the manufacturer.

Advantages: The advantages of a vortex meter are many. They are summarized below:

- No moving parts to wear
- No routine maintenance required
- Can be used for liquids, gases, and steam
- Stable long term accuracy and repeatability
- Lower cost of installation than traditional orifice-type meters
- Available in a wide variety of temperature ranges from -300F to roughly 800F
- Bar-like bluff design allows particulates to pass through without getting clogged
- Available for a wide variety of pipe sizes
- Available in a wide variety of communication protocols



Figure 6
This photo shows a typical vortex meter. It may be installed horizontally or vertically in the pipe.

Disadvantages: There are only a couple of things to watch out for when considering a vortex meter. First, they are not a good choice for very low fluid velocities, and therefore cannot be recommended below about 0.3 ft/sec. At this low flowrate, the vortices are not strong enough to be picked up accurately.

In addition to the above, be aware that a minimum length of straight-run pipe is required upstream and downstream of the meter for the accurate creation of vortices within the flowmeter. Ten pipe diameters before and after the point of installation are typically recommended, but the minimum length could be greater if there are elbows or valves nearby. This is only a disadvantage if the installation area does not allow for this straight run of pipe.

Applications: Vortex meters have become extremely popular in recent years and are used in a variety of applications and industries. Below is a summary of some of the main uses of a vortex meter.

- Custody transfer of natural gas metering
- Flow of liquid suspensions
- Higher viscosity fluids
- Cryogenic fluids
- Steam measurement
- General water applications
- Chilled and hot water
- Water/glycol mixtures
- Condensate measurement
- Potable water
- Ultrapure & de-ionized water
- Acids
- Solvents

Vortex meters are also used widely in the oil, gas, petrochemical, and pulp & paper industries.

The Magnetic Flowmeter

Design Overview: The basic design principle of the magnetic flowmeter (Figure 7) is derived from Faraday's law of induction, which states that the voltage generated in a closed circuit is directly proportional to the amount of magnetic flux that intersects the circuit at right angles.

In this design, magnets are positioned above and below the pipe to produce a magnetic flux (B) along the Y-axis. Because of the movement of conductive fluid, at right angles to this magnetic field and at a velocity V along the Z-axis, a potential is induced into the flow stream. The instantaneous voltage produced between the electrodes is proportional to the fluid flow through the pipe. For this design, one can rewrite Faraday's Law as follows:

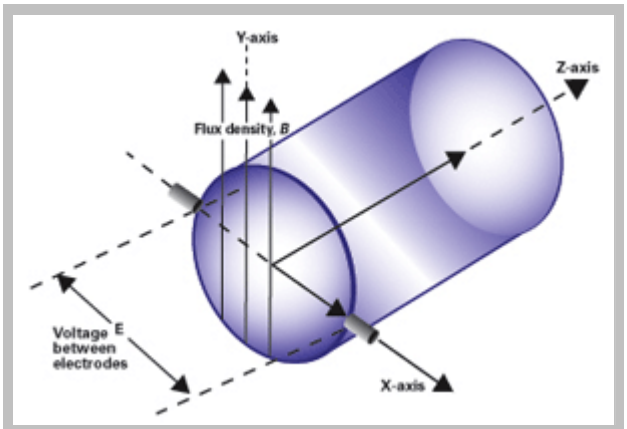


Figure 7
This illustration shows the principle of the manetic flowmeter. As magnetic flux is produced upward along the Y-axis, a voltage develops across the meter electrodes as conductive fluid moves through the pipe. The voltage signal is directly proportional to the fluid velocity.

E = kBdV

where,
E = the induced voltage between the sensing electrodes
k = a constant
B = the magnetic flux density
d = the distance between electrodes (equivalent to the pipe diameter)
V = the velocity of the fluid

Linear flow through a pipe can be expressed as the volumetric flowrate Q, divided by the cross-sectional area of the pipe A; therefore one can write

$V = Q/A = 4Q/\pi d^2$

Substituting this into the Faraday equation gives

$E = (4k/\pi d)BQ$

This can be solved for the volumetric flow rate Q, and leads to

$Q = (\pi d/4k)E/B$

This final equation shows that the volumetric flowrate Q is directly proportional to the induced voltage, E, between the electrodes.

There are two main methods of producing the magnetic flux density, B, across the pipe; alternating-current (a.c.) excitation, or pulsed, direct-current (d.c.) excitation.

In order to avoid past polarization problems encountered in a d.c.-excitation design, some magmeters use an a.c. excitation voltage. In this design, an a.c. voltage is used to create the magnetic field which, in turn, produces a varying-voltage signal across the electrodes. This is not a problem since the amplitude of the voltage, E, will still be proportional to the fluid velocity.

However, the development of some induction voltages across both the transformer coils and the electrodes is undesirable. For induction voltages that are 90 degrees out of phase with the signal voltage (called quadrature voltages), a phase-sensitive filtering circuit eliminates the unwanted voltage. Induction voltages that are in phase with the signal voltage can be eliminated with special zeroing procedures but this usually requires the fluid flow in the pipe to be fully stopped before zeroing; this may not be feasible in some applications.

Response time is quicker with a.c. excitation than with d.c. pulse-type units. This can be an advantage if the process flow changes quickly or contains hard particulate matter, like sludge, pulp-and-paper stock, mining slurries and polymers. Hard particulates impinging on the electrodes can generate signals that can be mistaken for noise as opposed to the actual flow signal. The 60-HZ sampling of the AC design will work very well in distinguishing between noise and actual flow signals in these types of applications. Outside of these more specialized cases

however, the d.c.-pulse design is more widely used since it eliminates many of the above-mentioned induction-voltages altogether.

In pulsed-d.c. excitation, the electromagnet coils are energized in short pulses or bursts. The electrode voltage is then measured before and after the d.c. excitation and the voltage difference is proportional to the flowrate. The advantage of the d.c. pulse design is that it eliminates the induction voltages described above, as well as the need to re-zero the meter at no flow conditions. Normally, the d.c. excitation is pulsed around 10 to 15 Hz. Some companies, in an effort to provide the advantages of the a.c. design, have increased the d.c. pulsing to 100 Hz. While this certainly allows the meter to handle more difficult flows, it may increase the amount of heat generated in the coils and can affect the lifetime of the instrument. Some new designs claim to minimize this heating effect.

As a final mention, it is worth noting that some magmeter designs have solved the problem of coating-type fluids leaving a non-conductive deposit on the meter electrodes. By embedding metal sheets in the magmeter lining, the electrodes no longer come in direct contact with the fluid, and the measured parameter becomes capacitance instead of voltage.



Figure 8
This photo shows a typical magnetic flowmeter, which can be installed horizontally or vertically in the pipe.

Advantages: The magmeter offers some very nice advantages. They are summarized below:

- Obstructionless flow
- Virtually no pressure drop
- Insensitivity to viscosity, specific gravity, temperature and pressure (within certain limitations)
- Will work with laminar, turbulent, and transitional flows
- Can respond well to fast changing flows (for high-frequency d.c.—pulse and a.c.—excitation designs only)
- Good accuracy (0.5 to 1%)
- No moving parts
- Can handle slurries and heavy particulates
- Lining protectors available for harsh, chemically corrosive, and abrasive fluids
- Inline and insertion designs available to handle pipe sizes from approximately 1/10" to 96"
- Available in a wide variety of communication protocols

Disadvantages: The only main disadvantage of the magmeter is that the fluid needs to be conductive. Therefore, liquids such as hydrocarbons and de-ionized water are not viable applications. The minimum required conductivity is normally in the range of 1-5 microSiemans/cm (mS/cm) but will vary from design to design. One manufacturer claims a minimum conductivity of 0.008 mS/cm while another recommends 20 mS/cm. Again, it is advisable to check with particular manufacturer's requirements.

The only other item to point out is that because this technology utilizes magnetic and electric fields, the pipe must normally be grounded. There are special grounding procedures that need to be followed for conductive piping; and for plastic pipes, special grounding rings must be used. Although this is technically not a disadvantage, it does add another step to the installation process and failure to properly ground the pipe can result in fluctuating flow signals.

Finally, it is not recommended to use graphite gaskets when installing a magmeter since the graphite could cause an electrically conductive layer to build up on the inside wall of the meter, causing erroneous signals. In the same spirit, it almost goes without saying that installation in an area containing stray electromagnetic or electrostatic fields is not recommended.

Applications: The magmeter can handle a wide variety of applications. Some of them are listed below:

- Water
- A variety of industrial effluents
- Paper pulp
- Mining slurries
- Brine
- Sludge
- Liquid food products
- Detergents
- Sewage
- Corrosive acids
- Solid bearing fluids
- Electrolytes
- Process chemicals

Problem liquids include petroleum products, crude oil, deionized water, and vegetable/animal fats.

Final Words

A word of caution: The technologies discussed within this article represent an overview of what is available on the market and the values in Table 1 are average values. While there are hundreds of different designs available, the purpose of this article is to give the reader enough knowledge to narrow down their application to one or two flowmeter technologies. For specific issues or additional design-parameters that should be considered, the manufacturers should be apprached.

Table 1: A Comparison of Flowmeter Parameters

Attribute	Bubble	Doppler	Transit-Time	Vortex	Magnetic
Gases	Yes	Yes[1]	Yes[1]	Yes	No
Steam	No	Yes[1]	Yes[1]	Yes	No
Liquids	Yes	Yes	Yes	Yes	Yes
Viscous liquids[2]	Yes	Yes	Yes	Yes	Yes
Corrosive liquids	Not recommended	Yes	Yes	Yes	Yes
Typical Accuracy	2%[3]	2%[4]	0.5%[4]	0.75-1.5%[5]	0.5-1%[5]
Typical Repeatability	1%[3]	0.5%[4]	0.2%[4]	0.2%[5]	0.2%[5]
Max pressure, psi	Vent[6]	N/A[7]	N/A[7]	300 to 400	600-800
Max temp., °F	212	N/A[7]	N/A[7]	400 to 500	250-300
Max pressure drop, psi	negligible	negligible	negligible	15 to 20	negligible
Typical turndown ratio[8]	300 to 1	50 to 1	N/A9	20 to 1	20 to 1
Average cost[10]	$600	$2,000 to $5,000	$5,000 to $8,000	$800 to $2,000	$2,000 to $3,000

1. While specialized Doppler and transit-time meters will work for gases and steam, they represent a small percentage of all Doppler and transit-time applications.
2. Upper viscosity limit will vary per manufacturer.
3. % of full-scale.
4. % of velocity.
5. % of flowrate.
6. Outlet must be vented to atmosphere
7. Non-contact device.
8. The turndown ratio is the ratio of maximum flow to minimum flow, also known as rangeability.
9. Transit-time technology can measure down to zero flow.
10. Cost values vary depending on process temperature and pressure, accuracy required and approvals needed.